



TECHNICAL WHITEPAPER

Industrial M.2 2280 NVMe SSD (DRAM Cache)

Industrial NVMe SSD with onboard DRAM cache, up to 8 TB capacity, sequential speeds $\geq 2900/2600$ MB/s, and endurance up to 3200 TBW for demanding workloads.

JEDEC Compliant

Industrial Temp
-40°C to 85°C

1-8TB

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Table of Contents

1. Executive Summary	1
2. Product Overview	1
Capacity Options	1
Key Attributes	1
Part Numbers (Industrial Grade)	2
3. Technology and Architecture	2
Controller and Data Path	2
NAND Flash	2
Onboard DRAM Cache	2
Error Correction and Monitoring	2
Power Management	2
Internal Block Diagram	3
4. Key Features and Differentiators	3
5. Technical Specifications	3
Physical Specifications	3
Power Supply	3
Power Consumption (Maximum Values, Reference Only)	3
Temperature Ranges	4
Endurance	4
Interface Pinout Summary (Power and Ground)	4
PCIe Interface Pins	4
6. Performance and Reliability	5
Sequential Transfer Speeds (CrystalDiskMark, Windows)	5
Random Performance	5
Latency	5
Reliability and Endurance	5
Environmental Restrictions	5

Quality Inspection	5
7. Applications and Target Markets	6
8. System Integration and Design Considerations	6
System Requirements	6
Power Supply Design	6
Thermal Management	6
Signal Integrity	6
Mechanical Integration	7
Storage and Handling	7
Notes on Unspecified Parameters	7
9. Standards Compliance and Quality	7
Compliance	7
Quality Assurance	7
Packaging and Delivery	8
10. Ordering Information	8
Standard Industrial- Grade Part Numbers	8
Contact Information	8
11. About Loongtion	8

1. Executive Summary

The Loongtion M.2 2280 NVMe SSD with DRAM Cache is an industrial- grade solid- state drive designed for embedded, industrial, medical, and commercial computing applications. Built around a China- domestic controller and Triple- Level Cell (TLC) NAND flash, the drive complies with the NVMe 1.4 protocol and operates over a PCIe 3.0 x4 interface via a standard M.2 M- Key connector.

Available in capacities from 1 TB to 8 TB, the drive delivers sequential read speeds of ≥2900 MB/s and sequential write speeds of ≥2600 MB/s (8 TB capacity, measured with CrystalDiskMark on Windows). Endurance is rated at up to 3200 TBW (8 TB) with 3000 P/E cycles, backed by a 5- year warranty. The industrial- grade variant supports an operating temperature range of –40 °C to +85 °C and a storage temperature range of –45 °C to +90 °C.

An onboard DRAM cache (approximately 1 GB per 1 TB of SSD capacity, sourced from Micron or ChangXin Memory Technologies (CXMT) depending on production batch) is included. The drive also supports TRIM, NCQ, S.M.A.R.T., ECC, wear leveling, and software- triggered secure erase.

This whitepaper provides a detailed technical description, including architecture, electrical specifications, performance data, reliability metrics, system integration guidance, and ordering information. All data are derived from the Loongtion consolidated technical fact sheet (Revision V1.1, 2024.02.01). Specifications are subject to change; users should always refer to the latest official datasheet for complete and current information.

2. Product Overview

The Loongtion M.2 2280 NVMe SSD (DRAM Cache) is a solid- state drive in the M.2 2280 form factor (22 mm × 80 mm, height ≤5 mm, weight <20 g). It connects to the host system through a standard M.2 M- Key 75- pin gold- finger connector with 3 μin gold plating. The interface implements NVMe 1.4 over a PCIe 3.0 x4 bus.

Capacity Options

Nominal Capacity	Formatted (Usable) Capacity
1 TB	953 GB
2 TB	1.86 TB
4 TB	3.64 TB
8 TB	7.44 TB

Note: Tolerance between nominal and usable capacity: ±10 %.

Key Attributes

- **Grade:** Industrial (I)
- **Controller:** China- domestic (specific model not specified in source documentation)
- **NAND Flash:** TLC (Triple- Level Cell), 3000 P/E cycles

- **DRAM Cache:** Onboard, ~1 GB per 1 TB of SSD capacity; suppliers: Micron or CXMT depending on production batch
- **Supported Features:** TRIM, NCQ, S.M.A.R.T., ECC, wear leveling, software- triggered secure erase (via pin 67), high- reliability power management with short- circuit and undervoltage protection
- **Warranty:** 5 years

Part Numbers (Industrial Grade)

Capacity	Part Number
1 TB	YIDJ01TNQT- S
2 TB	YIDJ02TNQT- S
4 TB	YIDJ04TNQT- S
8 TB	YIDJ08TNQT- S

3. Technology and Architecture

Controller and Data Path

The drive employs a China- domestic controller (specific model not disclosed in the source documentation) that manages all host- side NVMe protocol parsing, data verification, and NAND flash array management. Data flow follows this sequence:

1. The host communicates via the M.2 M- Key connector.
2. The controller parses NVMe commands and validates data on the link.
3. Valid data is then distributed across the TLC NAND flash array, which is organized as a multi- channel array (exact channel count not specified in source documentation).

NAND Flash

The NAND flash uses Triple- Level Cell (TLC) technology, rated for 3000 program/erase (P/E) cycles. Wear leveling is supported to extend the useful life of the NAND array across all P/E cycles.

Onboard DRAM Cache

An onboard DRAM cache (approximately 1 GB per 1 TB of SSD capacity) is populated using memory from either Micron or ChangXin Memory Technologies (CXMT), depending on production batch.

Error Correction and Monitoring

- **ECC:** Supported (specific algorithm not specified in source documentation).
- **S.M.A.R.T.:** Supported for self- monitoring, analysis, and reporting.
- **NCQ (Native Command Queuing):** Supported.

Power Management

The drive includes high- reliability power management circuitry with short- circuit and undervoltage protection. Power consumption varies by capacity and workload (see Section 5 for detailed values). Sleep power is consistent at 0.4 W across all capacities.

Internal Block Diagram

A block diagram is referenced as Figure 1 in the source documentation but is not reproduced in this whitepaper.

4. Key Features and Differentiators

- **Industrial- Grade Reliability:** Operating temperature range -40°C to $+85^{\circ}\text{C}$; storage temperature -45°C to $+90^{\circ}\text{C}$; 5- year warranty; up to 3200 TBW (8 TB).
- **Onboard DRAM Cache:** Provides the performance benefits typical of DRAM- based NVMe SSDs.
- **China- Domestic Supply Chain:** The controller and DRAM cache (when sourced from CXMT) are domestic; TLC NAND with 3000 P/E cycles.
- **High Sequential Performance:** Up to ≥ 2900 MB/s read, ≥ 2600 MB/s write (8 TB) over PCIe 3.0 x4.
- **Wide Capacity Range:** 1 TB to 8 TB in a single M.2 2280 module.
- **Software- Triggered Secure Erase:** Dedicated pin (pin 67) enables secure erase commands.
- **Power Management:** Consistent 0.4 W sleep power; short- circuit and undervoltage protection.
- **Standard Compliance:** NVMe 1.4, PCIe 3.0, M.2 M- Key mechanical/electrical.
- **Quality Packaging:** Anti- static bags, clean/dry packaging, certificate of conformance included.

5. Technical Specifications

Physical Specifications

Parameter	Specification
Form factor	M.2 2280
Dimensions	80 mm (± 0.2) $\times$ 22 mm (± 0.2)
Height	≤ 5 mm
Weight	< 20 g
Connector	M.2 M- Key, 75 pins
Gold plating thickness	3 μm

Power Supply

Parameter	Specification
Supply voltage	+3.3 V
Voltage range	3.15 V – 3.45 V
Ripple/noise (p- p)	≤ 50 mV

Power Consumption (Maximum Values, Reference Only)

Capacity	Sleep	Idle	Max Seq Read	Max Seq Write
1 TB	0.4 W	0.5 W	1.5 W	2.4 W
2 TB	0.4 W	0.6 W	3.5 W	3.7 W
4 TB	0.4 W	0.8 W	5.4 W	5.5 W
8 TB	0.4 W	0.8 W	7.8 W	8.8 W

Important: Power consumption data are for reference only. Actual values may vary by system configuration and workload.

Temperature Ranges

Parameter	Specification
Operating temperature	–40 °C to +85 °C
Storage temperature	–45 °C to +90 °C
Recommended storage area	0 °C to 35 °C, RH ≤80 %

Endurance

Capacity	TBW (Total Bytes Written)	P/E Cycles
1 TB	400 TBW	3000
2 TB	800 TBW	3000
4 TB	1600 TBW	3000
8 TB	3200 TBW	3000

Note: TBW tolerance is ±10 %.

Interface Pinout Summary (Power and Ground)

Pin Type	Pin Numbers
+3.3 V	2, 4, 12, 14, 16, 18, 70, 72, 74
Ground	1, 3, 9, 15, 21, 27, 33, 39, 45, 51, 57, 71, 73, 75

PCIe Interface Pins

Signal	Pin Assignment
PCIe Lane 3 Tx	PETn3 / PETp3
PCIe Lane 3 Rx	PERn3 / PERp3
PCIe Lane 2 Tx	PETn2 / PETp2
PCIe Lane 2 Rx	PERn2 / PERp2
PCIe Lane 1 Tx	PETn1 / PETp1
PCIe Lane 1 Rx	PERn1 / PERp1
PCIe Lane 0 Tx	PETn0 / PETp0
PCIe Lane 0 Rx	PERn0 / PERp0
Reference Clock	REFCLK_N (pin 53), REFCLK_P (pin 55)
PCIe Reset	PERST# (pin 50, active low)
PCIe Clock Req.	CLKREQ# (pin 52, active low)
Secure Erase	Pin 67 (labeled NC in table)

Note: Multiple pins are marked “No connection” (NC) or “Reserved” – see full pinout in the datasheet.

6. Performance and Reliability

Sequential Transfer Speeds (CrystalDiskMark, Windows)

Capacity	Average Sequential Read	Average Sequential Write
1 TB	≥2000 MB/s	≥1500 MB/s
2 TB	≥2300 MB/s	≥2000 MB/s
4 TB	≥2863 MB/s	≥2316 MB/s
8 TB	≥2900 MB/s	≥2600 MB/s

Note: Values are minimums (≥) verified with CrystalDiskMark on Windows. Actual performance may vary depending on system configuration, workload, and test environment.

Random Performance

- Not specified in source documentation.

Latency

- Not specified in source documentation.

Reliability and Endurance

- **TBW per capacity:** See Section 5.
- **P/E cycles:** 3000 (TLC NAND).
- **Warranty:** 5 years.
- **MTBF:** Not specified in source documentation.
- **Data retention:** Not specified.
- **Shock/vibration resistance:** Not specified.
- **Features supporting reliability:** ECC, wear leveling, S.M.A.R.T. monitoring, high- reliability power management with short- circuit and undervoltage protection.

Environmental Restrictions

- Do not store together with acids, alkalis, or corrosive substances.
- Transportation: ordinary vehicles, protect from rain and snow, handle with care.
- Storage area: temperature 0 °C to 35 °C, relative humidity ≤80 %, free from strong magnetic fields.

Quality Inspection

- **Appearance:** Housing surface free from burrs, deformation, contamination, corrosion; coating uniform; markings clear and legible.
- **Electrical performance:** SMART information check, file system data read/write test, normal temperature electrical performance test.
- Each drive is shipped with a test report and certificate of conformance.

7. Applications and Target Markets

The Loongtion M.2 2280 NVMe SSD (DRAM Cache) is designed for systems that require high-performance NVMe storage in challenging environments. Target application domains include:

- **Embedded Devices and Systems:** Industrial PCs, edge computing platforms, embedded controllers.
- **Industrial Equipment:** Factory automation, CNC machines, surveillance DVRs, test and measurement equipment.
- **Medical Equipment:** Patient monitors, diagnostic imaging systems, portable medical devices (operating temperature -40°C to $+85^{\circ}\text{C}$).
- **Commercial Computers:** Thin clients, workstations, servers requiring NVMe performance and reliability.

The drive is particularly suited for boot drives and data storage in systems with NVMe support where a wide temperature range, high endurance, and an onboard DRAM cache are required.

8. System Integration and Design Considerations

System Requirements

- The host hardware must support the NVMe protocol and provide an M.2 M- Key slot with PCIe x4 lanes.
- The operating system must have NVMe driver support.

Power Supply Design

- The SSD requires a +3.3 V supply within the range 3.15 V – 3.45 V with ripple $\leq 50\text{ mV p-p}$.
- Multiple power pins (nine total) distribute current; ensure adequate decoupling on the host board.
- Sleep power is consistent at 0.4 W across all capacities; idle and active power scale with capacity.
- The power management subsystem includes short-circuit and undervoltage protection – ensure the host power supply can handle peak current events.

Thermal Management

- Maximum power dissipation reaches 8.8 W (8 TB capacity under sequential write). The system thermal design should accommodate this, especially for sustained write operations. Use of heatsinks or forced airflow is recommended for high-capacity, high-write applications.
- The drive claims effective thermal dissipation; specific thermal throttling implementation details are not provided in the source documentation.

Signal Integrity

- PCIe differential pairs (PETn/p, PERn/p) require controlled-impedance routing on the host PCB.
- The reference clock (REFCLK_N, REFCLK_P) must be provided on pins 53 and 55 with PCIe-standard frequency.
- Reset (PERST#) and clock request (CLKREQ#) are active-low signals and must be driven by the host.
- The software-triggered secure erase pin (pin 67) can be employed for secure erase commands.

Mechanical Integration

- The M.2 2280 module measures 80 mm × 22 mm with a height ≤5 mm; ensure adequate clearance in the enclosure.
- Weight <20 g; no special mounting support needed.
- Connector gold plating is 3 μin; consider mating cycle limits for designs requiring frequent insertion/removal.

Storage and Handling

- Store at 0 °C to 35 °C, relative humidity ≤80 %, away from strong magnetic fields and corrosive substances.
- Product is shipped in anti-static bags with foam, certificate of conformance, and test reports.
- Do not stack heavy objects on the product; handle with care during installation.

Notes on Unspecified Parameters

- Power loss protection (PLP) details are not provided; consult the latest datasheet for any capacitor-based backup.
- Thermal throttling method is not described; it is advisable to monitor drive temperature via S.M.A.R.T. attributes.
- Over-provisioning configuration is not specified.
- Encryption support (AES, TCG Opal) is not confirmed in the source documentation.

9. Standards Compliance and Quality

Compliance

Standard / Specification	Status
NVMe 1.4	Compliant
PCIe 3.0	Interface compliant
M.2 M- Key (mechanical/electrical)	Compliant
JEDEC specific standards	Not specified in source
PCI-SIG certification	Not specified
RoHS	Not specified
CE / FCC	Not specified

Recommendation: For definitive compliance information, contact Loongtion directly or refer to the latest official product datasheet.

Quality Assurance

The product undergoes rigorous inspection and testing, including:

- **Appearance:** Burr-free housing, uniform coating, clear markings.
- **Electrical performance:** SMART information check, file system data read/write verification, normal temperature electrical performance test.

- Each drive is shipped with a test report and certificate of conformance.

Packaging and Delivery

- Packaged in anti- static bags.
- Outer carton includes anti- static foam, certificate of conformance, and corresponding test reports.
- Transportation may be via ordinary vehicles; protect from rain, snow, and corrosive substances.
- Storage conditions: 0 °C to 35 °C, RH ≤80 %, no strong magnetic fields, and not with acids/alkalis.

10. Ordering Information

Standard Industrial- Grade Part Numbers

Capacity	Part Number
1 TB	YIDJ01TNQT- S
2 TB	YIDJ02TNQT- S
4 TB	YIDJ04TNQT- S
8 TB	YIDJ08TNQT- S

Note: Part numbers for other grades or configurations are not provided in the source documentation. Contact Loongtion for additional options.

Contact Information

- **Company:** Ningbo Loongtion Intelligent Technology Co., Ltd.
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11. About Loongtion

Ningbo Loongtion Intelligent Technology Co., Ltd. (Loongtion®) is a developer and supplier of China- domestic memory and solid- state storage products. The company's portfolio includes:

- DDR3 / DDR4 / DDR5 / LPDDR4X SDRAM modules
- eMMC 5.1 embedded storage
- M.2 NVMe SSDs (including the product described in this whitepaper)
- NVMe BGA SSDs

Loongtion products are designed for industrial, embedded, medical, and commercial applications, offering customers reliable storage solutions with domestic supply- chain assurance.

For further technical support, sales inquiries, or to request the latest official datasheet, please contact:

- **Email:** hi@loongtion.com
- **Web:** www.loongtion.com

Disclaimer: This whitepaper is based on the Loongtion consolidated technical fact sheet (Revision V1.1, 2024.02.01). Specifications, including but not limited to performance, power consumption, endurance, and physical dimensions, are subject to change without notice. Parameters not explicitly stated herein (e.g., random IOPS, latency, MTBF, data retention, shock/vibration ratings, regulatory certifications) are not provided in the source documentation and should be verified with the latest official product datasheet or by contacting Loongtion directly.